

Real-Time Prediction of 30-Day Hospital Readmission Risk Using EHR Data

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ABSTRACT

Accurately predicting 30-day hospital readmissions continues to be a priority for healthcare organizations seeking to improve care quality and reduce costs. Despite advancements in healthcare analytics, this challenge remains complex due to the multifactorial nature of readmission risk. Early identification of high-risk patients prior to discharge enables targeted interventions when they are most likely to be effective. This study evaluates the use of real-time patient-level electronic health record (EHR) data to predict 30-day readmissions and compares the performance of logistic regression and machine learning models. A retrospective dataset of 26,014 inpatient admissions was analyzed with predictive models developed using clinical, utilization, and hospitalization-related variables available during the inpatient stay. Model performance was assessed using area under the receiver operating characteristic curve (AUROC), sensitivity, specificity, and precision. The logistic regression model achieved an AUROC of 0.747, while the machine learning models achieved AUROC values of 0.749 (Random Forest) and 0.756 (Gradient Boosting). Although the machine learning models demonstrated slightly higher performance, DeLong's test indicated that these differences were not statistically significant. Variables related to prior healthcare utilization, discharge disposition, and patient risk scores emerged as key predictors across all models. Overall, these findings suggest that despite modest improvements in predictive performance with machine learning approaches, logistic regression provides a practical and interpretable solution for clinical implementation, supporting the use of predictive modeling for the reduction of avoidable readmissions.

INTRODUCTION

Hospital readmissions within 30 days of discharge represent a persistent and costly challenge for modern healthcare organizations. Driven in part by financial penalties under value-based reimbursement models, these readmissions are widely used as a key quality metric and are closely monitored by hospitals, insurers, and regulatory bodies in the United States. Elevated readmission rates are associated with gaps in care coordination, insufficient discharge planning, and unmet post-discharge needs, making the reduction of avoidable readmissions a central focus of quality improvement initiatives (Sharda et al., 2025). As a result, healthcare organizations are increasingly seeking data-driven approaches to identify high-risk patients and implement targeted interventions prior to discharge.

Advancements in predictive analytics, coupled with the widespread adoption of electronic health record (EHR) systems, have created new opportunities to address this challenge. Although originally designed for documentation and record-keeping, EHR systems capture detailed patient-level information – including demographic characteristics, medical history, and clinical data – enabling a more comprehensive risk assessment (Tang et al., 2024). Prior research has identified several key predictors of hospital readmissions, including length of stay, comorbidities, disease complexity indicators, and prior healthcare utilization patterns, all of which reflect potential underlying patient complexity and care coordination challenges (Selmer et al., 2025; Li et al., 2024).

To leverage these data, various predictive modeling techniques have been applied, from traditional statistical approaches such as logistic regression (LR) to more advanced machine learning (ML) methods. While ML models have demonstrated improvements in predictive accuracy, they often do so at the cost of interpretability and transparency, which are critical considerations in clinical settings (Linardatos, et al., 2020). Additionally, many prior studies rely on retrospective, post-discharge datasets, limiting their applicability to real-time clinical decision support.

This study aims to evaluate the effectiveness of real-time predictive modeling using patient-level EHR data to identify patients at high risk for 30-day readmission prior to discharge. In addition, this research compares the performance of an LR model with ML approaches to examine the tradeoff between predictive accuracy and model complexity. Given the importance of interpretability and its association with clinical adoption (Vani et al., 2025), the practical implications of model selection are also considered. By leveraging predictive analytics and variables available during hospitalization, this research seeks to contribute to the development of decision support tools that support care coordination, improve patient outcomes, and reduce avoidable hospital readmissions.

OBJECTIVES

The primary objective of this study is to evaluate whether patient-level EHR data available during discharge planning can be used to accurately predict 30-day hospital readmissions. Specifically, this study aims to develop an LR model capable of identifying patients at elevated risk using variables accessible prior to discharge, thereby supporting earlier interventions and improved care coordination.

A secondary objective is to compare the performance of LR, a traditional statistical technique, with more advanced ML approaches, including Random Forest and Gradient Boosting, to determine whether increased model complexity results in meaningful improvements in predictive accuracy. Model

performance will be evaluated using standard classification metrics, including area under the receiver operating characteristic curve (AUROC), sensitivity, specificity, and precision.

In addition to predictive performance, this study examines the practical implications of model selection in healthcare settings. While ML models may offer improved accuracy, their increased complexity can limit transparency and hinder clinical adoption (Vani et al., 2025). In contrast, LR models provide more interpretable outputs that may be better suited for clinical decision support. As such, this study considers the trade-off between predictive accuracy and interpretability when evaluating the performance and usability of these approaches in real-time clinical workflows.

OVERVIEW OF STUDY

This study utilizes a retrospective dataset consisting of 26,014 inpatient admissions, where each observation represents a unique index admission that is eligible for a 30-day readmission outcome. The dataset includes a comprehensive set of variables derived from an EHR system, encompassing demographic, clinical, utilization, and hospitalization-related factors. All variables were intentionally selected based on their availability during the inpatient stay, ensuring that risk predictions could be generated prior to discharge.

The primary outcome variable is a binary indicator of whether a patient was readmitted within 30 days of discharge. Predictor variables include patient age, gender, admission diagnoses, length of stay, discharge disposition, prior healthcare utilization indicators, and risk scores. These variables were selected based on prior research identifying them as key predictors of readmission risk, particularly those reflecting clinical severity, length of stay, and utilization patterns (Selmer et al., 2025; Li et al., 2024).

A comparative modeling approach is used to evaluate both LR and ML techniques. Model performance is assessed using standard classification metrics, including AUROC, sensitivity, specificity, and precision. This approach aligns with prior studies that employ consistent evaluation metrics to enable meaningful comparisons across predictive models (Sharda et al., 2025; Pandey et al., 2025).

RESEARCH QUESTIONS AND HYPOTHESES

Research Questions

The overarching aim of this study is to enable earlier identification of patients at high risk of readmission during their inpatient stay, thereby supporting improvements in discharge planning and care coordination. Prior research has shown that 30-day hospital readmissions are associated with worsened clinical outcomes and increased healthcare costs (Sharda et al., 2025), underscoring the need for effective risk-stratification tools. Guided by this motivation, the primary research question is:

RQ1. Can patient-level clinical and demographic data available during hospitalization be used to accurately predict 30-day hospital readmissions?

Building on this, a secondary research question examines how different modeling approaches impact predictive performance and interpretability:

RQ2. How does the performance of a traditional statistical model, such as LR, compare to ML models, including Random Forest and Gradient Boosting, in predicting 30-day readmission risk?

Overview of Hypotheses

The research questions are addressed through two primary hypotheses. The first hypothesis (H1) evaluates whether patient-level EHR data can be used within an LR model to provide meaningful predictive capability for 30-day readmissions prior to discharge. The second hypothesis (H2) examines whether increased model complexity through ML techniques results in improved predictive performance relative to LR. Together, these hypotheses support both the predictive capability and comparative objectives of the study.

H1: Prediction Capability Hypothesis

The primary research question examines whether patient-level clinical and demographic data available during hospitalization can be used to accurately predict 30-day readmissions.

- **H_{0,1} (Null Hypothesis):** Patient-level clinical and demographic variables available during hospitalization are not significantly associated with 30-day readmission risk and do not meaningfully predict readmissions.
- **H_{1,1} (Alternative Hypothesis):** Patient-level clinical and demographic variables available during hospitalization are significantly associated with 30-day readmission risk and can be used to meaningfully predict readmissions.

Based on prior research reporting AUROC values ranging from 0.60 to 0.70 for LR models in 30-day readmission prediction (Pandey et al., 2025; Sharda et al., 2025), this study defines meaningful predictive performance as an AUROC of 0.70 or greater for H1 evaluation.

H2: Model Comparison Hypothesis

The secondary research question evaluates whether ML models provide improved predictive performance compared to traditional statistical approaches.

- **H_{0,2} (Null Hypothesis):** There is no statistically significant difference in performance between LR and ML models in predicting 30-day readmission risk.
- **H_{1,2} (Alternative Hypothesis):** ML models provide a statistically significant improvement in performance than LR in predicting 30-day readmission risk.

For H2 testing, DeLong's test for correlated receiver operating characteristic (ROC) curves will be used to assess whether each ML model achieves a statistically significant increase in AUROC relative to the LR model.

LITERATURE REVIEW

Hospital readmissions within 30 days of discharge continue to represent a significant challenge for U.S. healthcare systems, as they are associated with increased costs, reduced quality of care, and financial penalties under value-based care models. As a result, there has been growing interest in the development of predictive models capable of identifying patients at high-risk of readmission during hospitalization. The widespread adoption of EHR systems has further enabled this effort by providing

researchers with access to detailed, real-world clinical data. This literature review examines recent research on (1) the use of EHR data for predictive modeling, (2) key predictors of hospital readmissions, (3) machine learning applications, and (4) the trade-offs between predictive performance and interpretability. Together, these themes provide a foundation for evaluating predictive models of 30-day readmission risk.

Predictive Modeling with EHR Data

The expansion of EHR systems in recent decades has created significant opportunities for healthcare research. Although initially designed for documentation purposes, EHR data captures a wide range of patient information, including demographics, diagnoses, physiological indicators, and treatments. As highlighted by Tang et al. (2024), EHR-based datasets enable the study of real-world patient populations and outcomes, positioning them as valuable resources for clinical risk modeling. With awareness and careful consideration of data quality limitations, these datasets can be effectively used to support predictive analytics.

Data quality remains a critical concern when working with EHR data. As emphasized by Goldstein et al. (2022), EHRs were not designed for research and contain large volumes of user-generated data that may include inaccuracies, misclassifications, and missing values. If not properly addressed, these limitations can introduce bias into predictive models. As the authors note, however, discarding imperfect data may unnecessarily reduce statistical power and limit the usefulness of analyses. As such, careful preprocessing and validation are essential to ensure the resulting analyses are reliable and trustworthy in healthcare settings. Despite these challenges, EHR data has become indispensable in modern healthcare analytics due to its widespread availability and depth.

Key Predictors of Hospital Readmissions

Prior research has consistently identified clinical factors, patient demographic characteristics, and healthcare utilization patterns as key predictors of hospital readmissions. Studies by Li et al. (2024) and Selmer et al. (2025) demonstrate that clinical indicators such as comorbidities, disease complexity, and length of stay are strongly associated with increased readmission risk. Additionally, demographic characteristics – including age, gender, race, and ethnicity – can provide important context for understanding disparities in outcomes. As Li et al. (2024) point out, however, some demographic variables may not meaningfully improve predictive models using general population data, as their effects can be highly localized.

Healthcare utilization patterns are particularly strong predictors of 30-day readmissions. Patients with frequent emergency department visits or prior hospital admissions tend to have higher readmission rates, often due to underlying chronic conditions and associated challenges with care coordination (Selmer et al., 2025). These patterns can signal instability in a patient's health status and highlight the need for enhanced post-discharge support.

Machine Learning Applications in Readmission Prediction

Advanced modeling techniques, such as ML, have shown an enhanced ability to predict hospital readmissions. LR remains one of the most commonly used approaches in the prediction of readmission risk; however, these models typically demonstrate only modest predictive performance with area under the curve (AUC) values between 0.60 and 0.70 (Pandey et al., 2025; Sharda et al., 2025). In contrast, ML

techniques such as Random Forest and Gradient Boosting are better suited to capturing complex, non-linear relationships common to healthcare data, potentially providing improved predictive accuracy.

Interpretability and Practical Considerations

While ML models may offer improved predictive performance, these gains often come at the cost of increased complexity. As Linardatos et al. (2020) emphasize, many ML models function as “black boxes,” offering limited transparency into how predictions are generated. This lack of interpretability can be problematic in healthcare settings where clinicians must be able to understand and trust model outputs. Alternatively, traditional statistical models such as LR offer greater transparency through interpretable coefficients and odds ratios. This creates an important trade-off between predictive accuracy and interpretability, which must be carefully considered when implementing predictive models in clinical practice.

Summary of the Literature Review

Existing literature demonstrates that 30-day readmissions remain a complex and persistent challenge, while also highlighting the potential of EHR data as a foundation for predictive modeling. Clinical indicators, patient demographics, and healthcare utilization patterns have been consistently identified as key predictors of readmission risk. Although traditional statistical models have been widely used in this domain, recent advancements in ML offer opportunities for improved predictive performance. However, challenges related to data quality, bias, and model interpretability remain important considerations. Collectively, this body of research underscores the need to balance predictive accuracy with transparency, providing a strong foundation for the development and evaluation of models aimed at identifying patients at high risk of readmission and supporting clinical decision-making.

RESEARCH DESIGN

Methodology

This study employs a quantitative observation research methodology to evaluate the predictive capability of EHR data using an LR model and to compare the performance of LR and ML models in predicting 30-day hospital readmissions. A quantitative approach is appropriate given the use of structured numerical and binary-encoded categorical variables to model the relationships between predictor variables and a binary readmission outcome.

The research design is retrospective, utilizing historical inpatient admission data from calendar year 2025 to train and test the predictive models. This design enables the analysis of real-world clinical data without manipulation of variables, which is essential for maintaining observational validity in healthcare research. As noted by Tang et al. (2024), EHR data provides a robust foundation for predictive modeling, particularly when potential challenges such as clinical behavior bias and missing data are appropriately addressed.

The methodology incorporates a predictive analytics framework with an emphasis on model performance and generalizability. To support this, the dataset is partitioned into training and testing subsets, allowing for the evaluation of model performance on unseen data. This approach reduces the risk of overfitting and aligns with established best practices in healthcare analytics (Adhiya et al., 2024; Pandey et al., 2025).

Finally, this methodology is designed to address both research questions. RQ1 evaluates whether patient-level EHR data can meaningfully predict hospital readmissions, while RQ2 examines differences in predictive performance between LR and ML models. By integrating predictive modeling with comparative evaluation, this approach enables a comprehensive assessment of both model effectiveness and practical applicability in healthcare settings.

Methods

Data Preprocessing and Feature Engineering

The data preprocessing stage involved several steps to prepare the dataset for modeling. First, 30-day readmission outcomes were captured by identifying cases in which an admission date occurred within 30 days of a discharge date for patients with matching identifiers. Observations that did not meet study inclusion criteria were then removed, including cases without a complete 30-day follow-up period. This ensured that all observations were eligible for an outcome assessment.

Next, variables not available prior to discharge were excluded to align with the study's objective of pre-discharge prediction. All remaining variables were confirmed to be available within the EHR system in real time. Additionally, selected categorical variables (e.g., discharge disposition, insurance coverage) were consolidated into fewer categories to reduce sparsity and improve model stability, particularly for low-frequency classes.

Missing values were addressed using techniques appropriate to each variable's type and distribution. In some cases, missingness was retained by creating a "missing" category prior to encoding. Most missing values were associated with patients lacking sufficient information to calculate risk scores. To better capture non-linear relationships and mitigate the impact of missingness, numeric risk scores were transformed into multi-class categorical variables.

Categorical variables were then standardized for modeling. Dichotomous variables were converted into binary indicators, while multi-class variables were transformed using one-hot encoding, a process that creates a separate binary variable for each mutually exclusive category except for a designated reference group that is omitted from the dataset. These transformations ensured consistent representation of categorical data across modeling techniques. Finally, all personally identifiable information was removed prior to modeling to ensure that only de-identified records were used in the analysis.

Model Development

To operationalize the methodology, this study employs both LR and ML models, including Random Forest and Gradient Boosting, within a Python-based analytical environment. The LR model serves as the baseline model due to its interpretability and widespread use in healthcare applications. It estimates the probability of 30-day readmission using coefficients that describe the direction and magnitude of each predictor's relationship with the outcome.

The ML models are included to evaluate whether increased model complexity leads to improved predictive performance. Random Forest and Gradient Boosting were selected based on their ability to reduce overfitting and capture non-linear relationships within complex datasets (Hidayaturrohman & Hanada, 2024). Prior research suggests that these techniques may outperform traditional statistical

models such as LR when applied to healthcare data characterized by high dimensionality and variable interactions (Pandey et al., 2025; Sharda et al., 2025).

Handling Class Imbalance

The final dataset contains 26,014 observations, of which 2,413 represent a positive outcome (30-day readmission), corresponding to approximately 9.3% of the sample. This distribution reflects a substantial class imbalance, which poses challenges for predictive modeling because models may become biased toward the majority class and fail to reliably identify high-risk patients.

To address this, class weighting techniques are applied during model training for both the LR and Random Forest models, increasing the relative importance of the minority class. For the Gradient Boosting model, a scaling parameter is used to achieve a similar adjustment. These approaches enable the models to account for class imbalance without altering the underlying data distribution, preserving the integrity of the dataset.

Model Evaluation

Model performance is evaluated using a consistent set of standard classification metrics to ensure comparability across modeling approaches. The primary evaluation metric is the area under the receiver operating characteristic curve (AUROC), which measures the model's ability to correctly discriminate between readmitted and non-readmitted patients across classification thresholds.

Additional evaluation metrics include:

- Sensitivity (recall): the proportion of correctly identified readmitted patients
- Specificity: the proportion of correctly identified non-readmitted patients
- Precision: the proportion of predicted readmissions that are correctly classified

Beyond AUROC, particular emphasis will be placed on the sensitivity measure, as correctly identifying high-risk patients is critical for clinical decision-making and resource allocation. While false positives may result in precautionary interventions, failing to identify truly high-risk patients may lead to missed opportunities for prevention and increased downstream healthcare costs.

This evaluation framework aligns with prior research on hospital readmission prediction and supports consistent comparison across modeling techniques (Pandey et al., 2025; Sharda et al., 2025).

Analytical Tools and Implementation

Due to system access limitations associated with extracting data from the EHR system, all data preprocessing was conducted in Microsoft Excel. This approach was necessary due to constraints within the healthcare data environment, where integration with external analytical tools is restricted. Following de-identification, the dataset was transferred outside the secure environment in full compliance with data governance policies.

Subsequently, the cleaned dataset was imported into a Python-based analytical environment, where all analyses were performed. Standard data science libraries were utilized to support data manipulation, model development, and evaluation, including:

- pandas for data manipulation,

- NumPy for numerical operations,
- scikit-learn for model development and evaluation,
- And XGBoost for Gradient Boosting modeling.

Limitations

Several limitations should be considered when interpreting the results of this study. Most notably, the dataset captures readmissions within a two-hospital system, including only patients who were readmitted to one of two hospitals from which they were discharged during the index admission. Readmissions to external hospitals, as well as patients who were admitted to the study hospitals after being discharged elsewhere, were not captured. These omissions may limit the completeness of the outcome variable and affect the generalizability of the model.

Additionally, as all observations originate from a single regional health system, localized clinical practices, patient demographics, and operational workflows may further limit generalizability. These local effects may influence both the predictors and the readmission patterns in the dataset, potentially limiting the model's applicability to other healthcare settings.

The study also relies on retrospective EHR data, which may contain inaccuracies, missing values, and inconsistencies due to variations in clinical documentation practices (Goldstein et al., 2022). Although data preprocessing steps were implemented to address these issues, residual data quality concerns may still impact model performance. Furthermore, the dataset primarily consists of structured variables. The inclusion of unstructured data, such as provider notes, has been shown to improve predictive performance in prior research (Hidayaturrohmah & Hanada, 2024; Pandey et al., 2025).

Lastly, an important limitation relates to the use of length of stay and discharge disposition as predictor variables. Because the dataset is retrospective, these variables reflect actual observed values. In a real-time deployment, where the model is intended to operate, these variables would instead be replaced by expected values, which are subject to change during hospitalization. Although these estimates become more accurate as discharge approaches, their variability may result in slightly lower predictive performance in real-world applications compared to the results observed in this study.

Ethical Considerations

A primary ethical consideration in this study is the use of EHR data for predictive modeling, which requires careful attention to patient privacy and data security. The dataset was de-identified in compliance with the Safe Harbor Method of the Health Insurance Portability and Accountability Act (HIPAA), ensuring that no personally identifiable information was accessible during the modeling process. Due to organizational constraints, data preprocessing was conducted within a controlled environment prior to transfer for analysis. These measures were taken to protect patient confidentiality and minimize the risk of unauthorized data exposure.

Predictive modeling in healthcare also raises important concerns related to bias, fairness, and equity. Variables such as race, ethnicity, and insurance status may reflect underlying disparities in healthcare access and outcomes. While these variables can enhance predictive accuracy, their inclusion risks reinforcing existing inequities in care if model outputs are not interpreted and applied carefully. Out of precaution, race, ethnicity, and insurance status variables were removed from the modeling dataset.

Transparency and interpretability represent additional ethical considerations in the deployment of predictive modeling in healthcare settings. Although ML models may offer improved predictive performance, their limited interpretability can make it difficult for clinicians to understand how predictions are generated (Linardatos et al., 2020). In contrast, LR provides more transparent and explainable outputs, fostering greater trust and adoption in clinical settings. Importantly, the models in this study are intended to support – rather than replace – clinical judgment, ensuring that healthcare providers retain full authority over decision-making.

FINDINGS

Analytical Dataset Description

The dataset used in this study included 26,014 inpatient admissions, of which 2,413 resulted in a 30-day readmission as captured by the outcome variable. This corresponds to a readmission rate of approximately 9.3%, reflecting a significant class imbalance in which more than 90% of encounters represent non-readmitted patients. Following data preprocessing and feature selection, 26 predictor variables and one outcome variable (readmit_30d) remained, for a total of 27 variables, as displayed in Figure 1.

Figure 1

Structure of the Modeling Dataset

#	Column	Non-Null Count	Dtype
0	age	26014 non-null	int64
1	gender_female	26014 non-null	int64
2	dispo_post_acute	26014 non-null	int64
3	dispo_transfer	26014 non-null	int64
4	dispo_ama	26014 non-null	int64
5	los_days_capped	26014 non-null	int64
6	infection_flag	26014 non-null	int64
7	adm_dx_sepsis	26014 non-null	int64
8	adm_dx_pna	26014 non-null	int64
9	adm_dx_hf	26014 non-null	int64
10	adm_dx_copd	26014 non-null	int64
11	adm_dx_rf	26014 non-null	int64
12	adm_dx_aki	26014 non-null	int64
13	mort_risk_medium	26014 non-null	int64
14	mort_risk_high	26014 non-null	int64
15	mort_risk_missing	26014 non-null	int64
16	med_risk_medium	26014 non-null	int64
17	med_risk_high	26014 non-null	int64
18	ed_low	26014 non-null	int64
19	ed_moderate	26014 non-null	int64
20	ed_high	26014 non-null	int64
21	ip_admit_utilization_low	26014 non-null	int64
22	ip_admit_utilization_moderate	26014 non-null	int64
23	ip_admit_utilization_high	26014 non-null	int64
24	icu_admit_flag	26014 non-null	int64
25	rrt_flag	26014 non-null	int64
26	readmit_30d	26014 non-null	int64
dtypes: int64(27)			

Note. The dataset includes 26,014 rows and 27 variables. None of the variables contain null values, and all are of an integer data type.

Descriptive Statistics

Among the 27 total variables included in the analytical dataset, two are discrete numeric variables that represent underlying continuous constructs. The age variable (age) corresponds to patient age in years and ranges from 18 to 104, with a mean of 55.3. The length-of-stay variable (los_days_capped) measures admission duration in days and was truncated at a maximum of 30 days to reduce the influence of extreme outliers on model performance. This modified variable ranges from 0 to 30 days and has a mean of 3.78, indicating a right-skewed distribution. Additional descriptive statistics for these variables are presented in Figure 2.

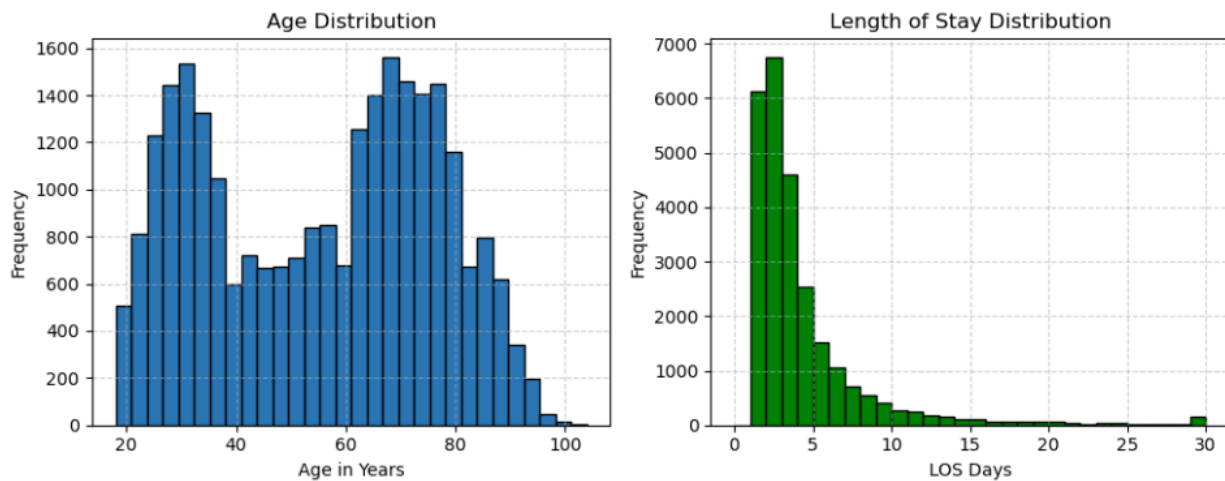
The distributions of the age and length-of-stay variables were further examined using histograms, as shown in Figure 3. Age demonstrates an irregular, multi-peaked distribution with concentrations among middle-aged and older adult populations. In contrast, length-of-stay exhibits a strong right skew, with the majority of admissions lasting five days or fewer and a long tail representing extended hospitalizations.

Figure 2
Descriptive Statistics for Age and Length-of-Stay Variables

	age	los_days_capped
count	26014.00	26014.00
mean	55.25	3.78
std	20.97	4.08
min	18.00	0.00
25%	35.00	2.00
50%	59.00	3.00
75%	73.00	4.00
max	104.00	30.00

Figure 3

Distributions of Age and Length-of-Stay Variables



Bivariate Analyses of Readmission Risk

Bivariate analyses were conducted to evaluate the relationship between the remaining 24 binary predictor variables and 30-day readmission. Figure 4 presents the results for the dichotomous binary variables representing patient gender and clinical severity indicators. Gender demonstrated only a modest association with readmission. Although female patients (`gender_female` = 1) were more highly represented in the dataset, higher readmission rates were observed among male patients (10.8% compared to 8.2%).

Figure 4

Bivariate Analyses of Gender and Clinical Severity Indicators to 30-Day Readmissions

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Gender	<code>gender_female</code>	15364	59.1	8.2
1	Gender	(Reference) <code>gender_male</code>	10650	40.9	10.8
2	Clinical Indicators	<code>infection_flag</code>	1720	6.6	19.2
3	Clinical Indicators	(Reference) <code>no_infection</code>	24294	93.4	8.6
4	Clinical Indicators	<code>icu_admit_flag</code>	3815	14.7	12.7
5	Clinical Indicators	(Reference) <code>no_icu_admit</code>	22199	85.3	8.7
6	Clinical Indicators	<code>rrt_flag</code>	2352	9.0	13.1
7	Clinical Indicators	(Reference) <code>no_rrt</code>	23662	91.0	8.9

The clinical indicator variables showed stronger associations. For example, patients with a positive infection indicator (`infection_flag` = 1) exhibited a readmission rate of 19.2%. Rapid Response Team (RRT) activation and ICU admission during hospitalization were also associated with increased readmission rates compared to their respective reference groups, although these differences were more modest.

Bivariate results for admission diagnosis variables are presented in Figure 5. These diagnoses were originally in a multi-select field and were converted into distinct binary indicator variables. Readmission rates across diagnoses – including sepsis, pneumonia, heart failure, COPD, respiratory failure, and acute kidney injury – were relatively similar, ranging from 11.3% for pneumonia (`adm_dx_pna`) to 14.9% for heart failure (`adm_dx_hf`).

Results for the remaining one-hot encoded variables are displayed in Figure 6. Discharge disposition demonstrated a strong association with readmission. Patients discharged to another facility (`dispo_transfer`), such as a skilled nursing or acute rehabilitation facility, exhibited a readmission rate of 49.8%, while those who left against medical advice (`dispo_ama`) had a readmission rate of 18.6%. Mortality risk scores showed a modest association, ranging from 6.7% among low risk patients to 18.7% among high risk patients. Medication risk scores demonstrated a similar pattern, increasing from 6.0% to 28.0% across categories. The largest differences were observed in prior healthcare utilization variables. Readmission rates increased substantially across levels of both ED utilization in the prior six months (6.7% to 28.6%) and lifetime inpatient utilization (5.5% to 38.5%).

Figure 5

Bivariate Analyses of Admission Diagnoses to 30-Day Readmissions

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Admission Diagnoses	<code>adm_dx_sepsis</code>	611	2.3	14.2
1	Admission Diagnoses	(Reference) <code>no_sepsis</code>	25403	97.7	9.2
2	Admission Diagnoses	<code>adm_dx_pna</code>	1114	4.3	11.3
3	Admission Diagnoses	(Reference) <code>no_pna</code>	24900	95.7	9.2
4	Admission Diagnoses	<code>adm_dx_hf</code>	739	2.8	14.9
5	Admission Diagnoses	(Reference) <code>no_hf</code>	25275	97.2	9.1
6	Admission Diagnoses	<code>adm_dx_copd</code>	472	1.8	14.6
7	Admission Diagnoses	(Reference) <code>no_copd</code>	25542	98.2	9.2
8	Admission Diagnoses	<code>adm_dx_rf</code>	1370	5.3	12.3
9	Admission Diagnoses	(Reference) <code>no_rf</code>	24644	94.7	9.1
10	Admission Diagnoses	<code>adm_dx_aki</code>	976	3.8	14.1
11	Admission Diagnoses	(Reference) <code>no_aki</code>	25038	96.2	9.1

Figure 6

Bivariate Analyses of One-Hot Encoded Variables to 30-Day Readmissions

	Variable Group	Category	Count	% of Total	Readmission Rate
0	Discharge Disposition	(Reference) dispo_home	21984	84.5	8.2
1	Discharge Disposition	dispo_post_acute	3382	13.0	12.2
2	Discharge Disposition	dispo_transfer	217	0.8	49.8
3	Discharge Disposition	dispo_ama	431	1.7	18.6
4	Mortality Risk	(Reference) mort_risk_low	15492	59.6	6.7
5	Mortality Risk	mort_risk_medium	3717	14.3	14.3
6	Mortality Risk	mort_risk_high	2566	9.9	18.7
7	Mortality Risk	mort_risk_missing	4239	16.3	8.8
8	Medication Risk	(Reference) med_risk_low	13937	53.6	6.0
9	Medication Risk	med_risk_medium	9790	37.6	9.6
10	Medication Risk	med_risk_high	2287	8.8	28.0
11	ED Utilization	(Reference) ed_none	14784	56.8	6.7
12	ED Utilization	ed_low	7654	29.4	8.8
13	ED Utilization	ed_moderate	2615	10.1	18.0
14	ED Utilization	ed_high	961	3.7	28.6
15	Inpatient Utilization	(Reference) ip_admit_utilization_none	6809	26.2	6.6
16	Inpatient Utilization	ip_admit_utilization_low	15068	57.9	5.5
17	Inpatient Utilization	ip_admit_utilization_moderate	3332	12.8	24.8
18	Inpatient Utilization	ip_admit_utilization_high	805	3.1	38.5

Overall, higher readmission rates in this dataset were observed among patients with greater prior healthcare utilization, more complex discharge disposition, and elevated risk scores. In contrast, demographic characteristics and individual admission diagnoses exhibited smaller differences across groups.

Assessment of Multicollinearity

Multicollinearity in the analytical dataset was assessed using both a correlation matrix and variance inflation factor (VIF). This statistical phenomenon can lead to unstable model estimates and reduced interpretability when predictor variables are highly correlated (Reddy & Naresh, 2026). As shown in Figure 7, the correlation matrix indicates primarily low to moderate relationships among variables. Some moderate correlations are observed within groups of one-hot encoded variables, such as ED utilization, inpatient utilization, and risk score categories. These relationships are expected due to the mutually

exclusive structure of encoded categories rather than true redundancy. Regardless, their magnitude does not suggest a significant concern.

To further evaluate multicollinearity, VIF values were calculated for each predictor variable, with the top ten values presented in Figure 8. The highest observed VIF is approximately 2.57, which is well below commonly accepted thresholds of concern (e.g., $VIF \geq 4$) per Reddy & Naresh (2026). Consistent with the correlation analysis, higher VIF values are primarily associated with the one-hot encoded variables, though all values remain within acceptable limits.

Figure 7
Correlation Matrix of Predictor Variables in the Analytical Dataset

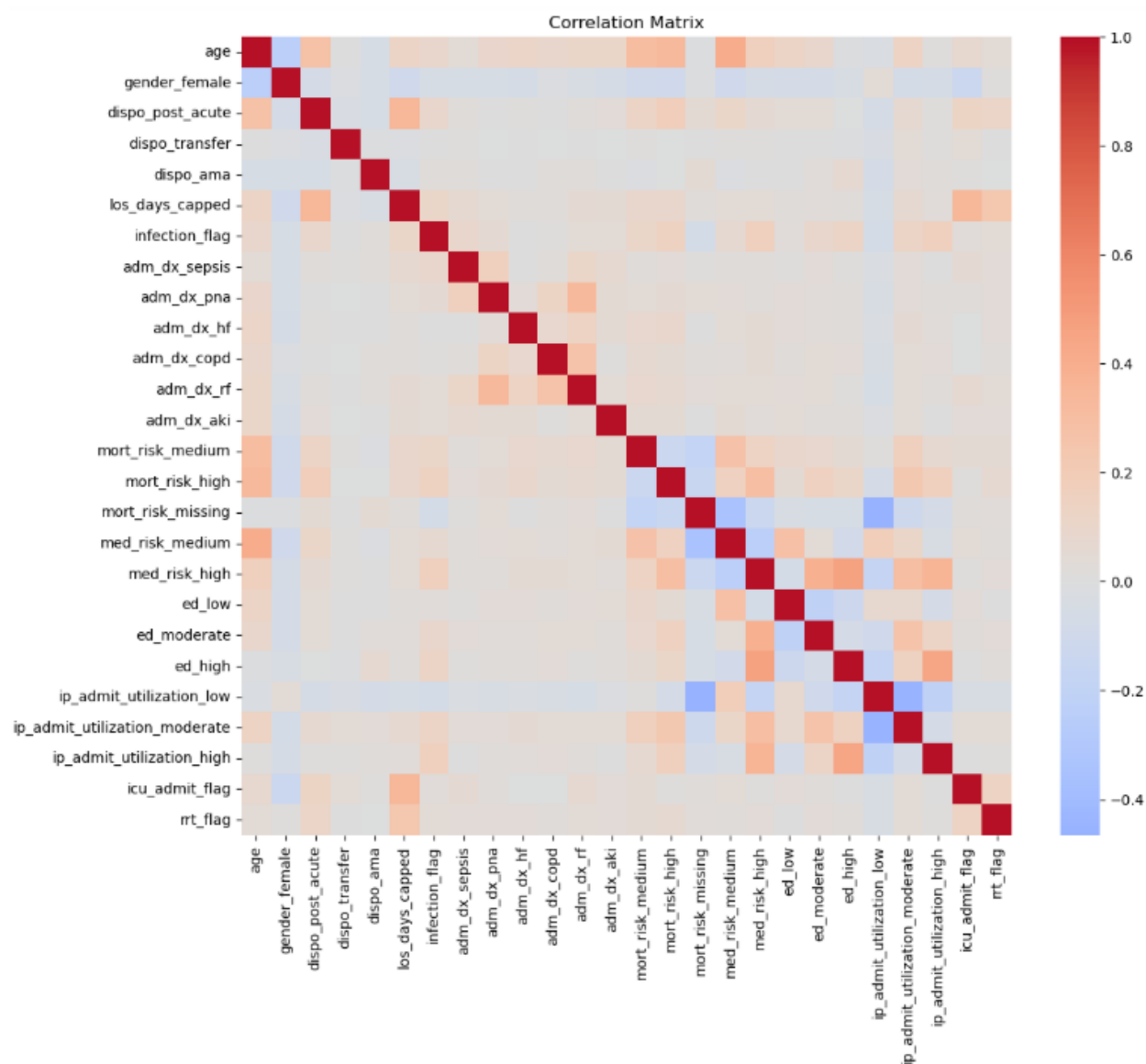


Figure 8

Top 10 Variance Inflation Factor (VIF) Values for Predictor Variables in the Analytical Dataset

	Feature	VIF
18	med_risk_high	2.565690
22	ip_admit_utilization_low	2.282701
17	med_risk_medium	2.114802
23	ip_admit_utilization_moderate	2.052613
16	mort_risk_missing	2.012789
1	age	1.808526
21	ed_high	1.794962
24	ip_admit_utilization_high	1.644281
20	ed_moderate	1.614120
15	mort_risk_high	1.500663

During an initial iteration of the multicollinearity assessment, a redundant missingness indicator (`med_risk_missing`) was identified as highly correlated with another missingness indicator (`mort_risk_missing`). This variable was removed from the dataset, and its information was appropriately consolidated into the reference category (`med_risk_low`). Following this adjustment, subsequent correlation and VIF analyses confirmed that multicollinearity was unlikely to have a meaningful impact on model stability or interpretation, supporting the inclusion of the remaining 26 predictor variables in the modeling phase.

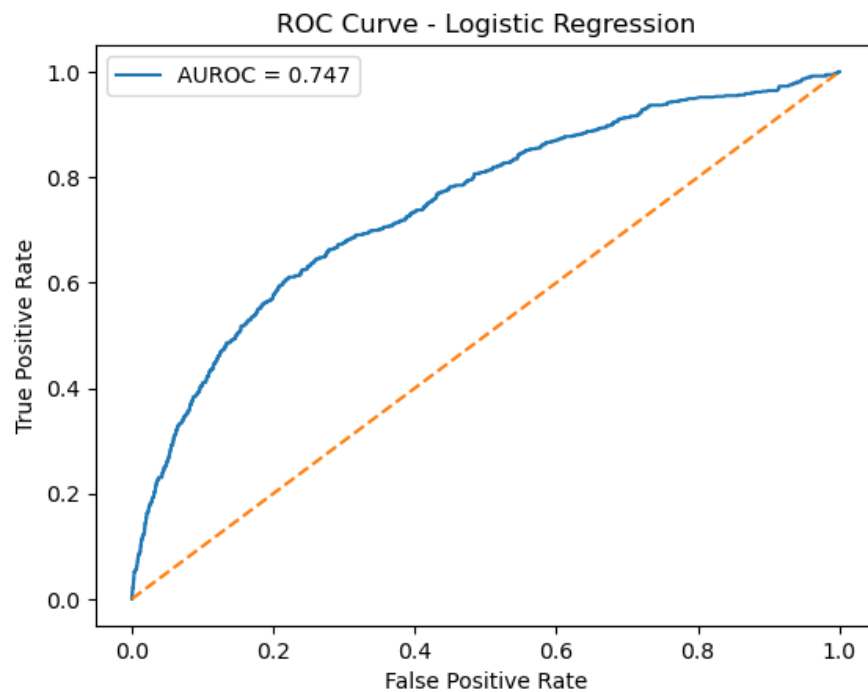
Logistic Regression Results

To evaluate the prediction capability hypothesis (H1), an LR model was developed to estimate 30-day readmission risk using patient-level demographic, clinical, and utilization variables. The EHR-based analytical dataset was partitioned into training (70%) and testing (30%) subsets, and class weighting was applied to address imbalance in the outcome variable.

The model achieved an AUROC score of 0.747, which indicates a useful discriminative ability in distinguishing between readmitted and non-readmitted patients. The corresponding ROC curve in Figure 9 demonstrates consistent performance across classification thresholds, with clear separation from the diagonal baseline.

Figure 9

ROC Curve for the Logistic Regression Model



Using the default classification threshold of 0.5, the confusion matrix for the testing dataset yielded 5,759 true negatives, 1,322 false positives, 318 false negatives, and 406 true positives. Based on these results, the model achieved a sensitivity (recall) of 0.56, a specificity of 0.81, and a precision value of 0.23. In other words, the LR model correctly identified 56% of readmitted patients and 81% of non-readmitted patients, while 23% of predicted readmissions were true positives. Lower precision values are expected in datasets with substantial class imbalance (Sharda et al., 2025).

To further assess model performance under alternative classification strategies, the threshold was reduced to 0.3 to prioritize sensitivity. At this threshold, sensitivity increased to 0.89, while specificity decreased to 0.35. Precision was also reduced to 0.12, reflecting a substantial increase in false positive predictions. This shift illustrates the trade-off between sensitivity and specificity, a common consideration in real-world clinical prediction applications, as noted by Sharda et al. (2025). In this context, the model casts a wider net to capture high-risk patients, while accepting a larger number of false positives. The consequences of misclassifying true readmissions as false negatives often outweigh the financial and operational burden of false positives in 30-day readmission prediction.

Analysis of model coefficients and corresponding odds ratios identified several influential predictors. Discharge disposition and prior inpatient utilization exhibited the strongest associations with readmission risk. Patients discharged to another facility demonstrated substantially higher odds of readmission compared to those discharged home. Similarly, higher levels of prior inpatient utilization were associated with increased readmission risk. In contrast, ED utilization variables demonstrated

negative coefficient values, suggesting a weaker or less direct relationship with readmission when considered alongside other predictors.

Machine Learning Results

To evaluate the model comparison hypothesis (H2), two ML models – a Random Forest and a Gradient Boosting (XGBoost) model – were developed using the same training and testing datasets as the LR model. The Random Forest model applied a consistent class-weighting approach as the LR model, while the Gradient Boosting model used a scaling parameter to address class imbalance.

The Random Forest model achieved an AUROC of 0.749, indicating performance comparable to the LR model. Using the default classification threshold of 0.5, a confusion matrix revealed 6,013 true negatives, 1,068 false positives, 363 false negatives, and 361 true positives. These results correspond to a sensitivity (recall) of 0.50, a specificity of 0.85, and a precision of 0.25 for the readmission class. Compared to LR, the Random Forest model produced fewer false positives but also identified fewer true readmissions.

The Gradient Boosting model achieved the highest performance with an AUROC of 0.756. The corresponding confusion matrix included 5,631 true negatives, 1,450 false positives, 295 false negatives, and 429 true positives. These results translate to a sensitivity (recall) of 0.59, a specificity of 0.80, and a precision of 0.23 for the readmission class. Relative to LR, the Gradient Boosting model demonstrated a modest increase in sensitivity while maintaining similar precision.

The ROC curves for both ML models are presented in Figure 10 and show similar overall performance and consistency across classification thresholds. Differences in AUROC between models are small, indicating comparable discriminative ability across modeling approaches.

Figure 10

ROC Curves for both machine learning models: Random Forest and Gradient Boosting

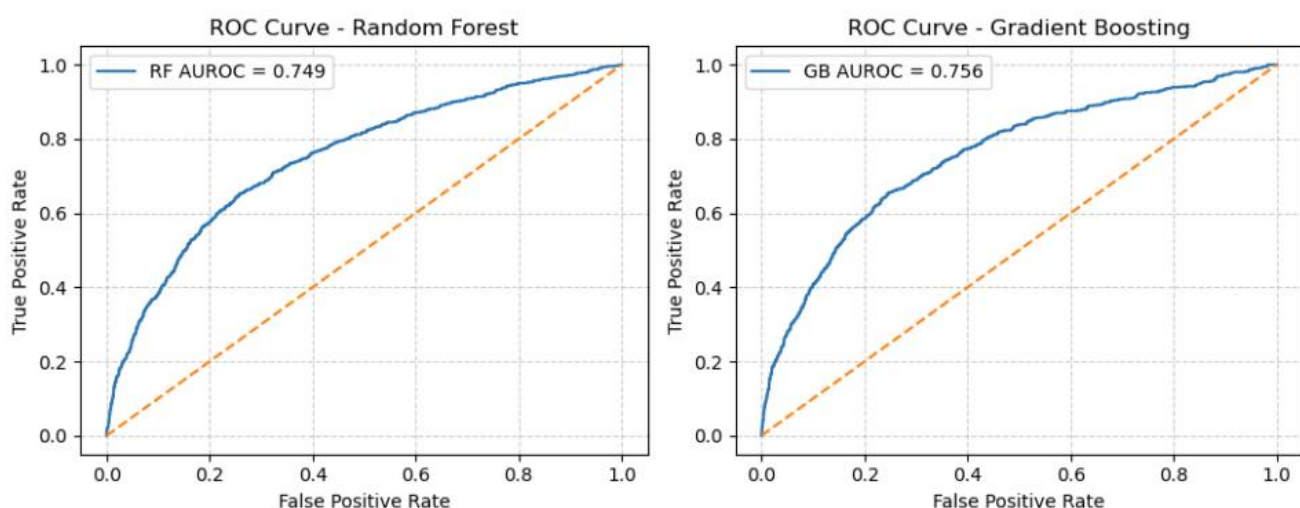


Table 1*Comparison of Model Performance*

Model	Logistic Regression	Random Forest	Gradient Boosting (XGBoost)
AUROC	0.747	0.749	0.756
Sensitivity (Recall)	0.56	0.50	0.59
Specificity	0.81	0.85	0.80
Precision	0.23	0.25	0.23

Model Comparison and Hypothesis Testing

To evaluate the comparative performance of modeling approaches, the LR model was assessed alongside two ML models: Random Forest and Gradient Boosting (XGBoost). Model performance metrics, including AUROC, sensitivity (recall), specificity, and precision, are presented in Table 1.

Predictive performance was similar across all models. The LR model achieved an AUROC of 0.747, while the Random Forest and Gradient Boosting models achieved AUROC values of 0.749 and 0.756, respectively. Although the Gradient Boosting model achieved the highest AUROC, the improvement over LR was modest.

In terms of classification performance, the models exhibited different trade-offs. For example, the LR model achieved a sensitivity of 0.56 and a specificity of 0.81, while the Random Forest model demonstrated lower sensitivity (0.50) but higher specificity (0.85), reflecting a more conservative classification pattern. The Gradient Boosting model not only achieved the highest AUROC but also the highest sensitivity at 0.59, indicating an improved ability to identify readmitted patients relative to LR, while maintaining similar precision.

To formally assess these differences, DeLong's test was used to evaluate the statistical significance of AUROC comparisons between models, in alignment with the model comparison hypothesis (H2). The comparison between LR and Random Forest yielded a p-value of 0.694, indicating no statistically significant difference in performance. The comparison between LR and Gradient Boosting resulted in a p-value of 0.073, suggesting a trend toward improved performance but not meeting the conventional threshold for statistical significance ($p < 0.05$). These results indicate that differences in AUROC across models are small and not statistically meaningful.

The results support the primary prediction capability hypothesis (H1), as the LR model demonstrated meaningful predictive performance ($\text{AUROC} \geq 0.70$), leading to rejection of the null hypothesis. However, the model comparison hypothesis (H2) is not supported. Neither ML model achieved a statistically significant improvement in AUROC relative to LR, and the null hypothesis for H2 cannot be rejected.

DISCUSSION

The findings of this study demonstrate that patient-level EHR data available during hospitalization can be used to meaningfully predict 30-day hospital readmissions. Predictive performance across all three models was consistent and aligned with ranges commonly reported in prior research. Notably, discharge disposition emerged as a strong predictor in this analysis, despite not being emphasized in the supporting literature. Variables reflecting prior healthcare utilization and patient risk scores were also among the most influential predictors, however, which is consistent with existing evidence (Li et al., 2024; Selmer et al., 2025). These factors often reflect underlying patient complexity and care coordination challenges. In contrast, individual admission diagnoses and demographic variables, including age and gender, demonstrated comparatively weaker associations, suggesting that readmission risk may be better captured with broader measures of utilization and severity.

Although ML models demonstrated slightly improved performance relative to LR, these gains were modest and did not reach statistical significance. The Gradient Boosting model achieved the highest AUROC (0.756); however, DeLong's test indicated that this improvement did not represent a meaningful increase in predictive capability. These findings suggest that increased model complexity does not necessarily translate into substantial performance gains, and that much of the predictive signal may already be captured by readily available EHR variables.

From a practical perspective, these results have important implications for healthcare organizations. LR offers strong predictive performance along with a high degree of interpretability. This transparency allows clinicians to understand how variables contribute to predicted risk, which is critical for building trust and supporting the implementation of predictive models in clinical workflows. While ML models are capable of capturing non-linear relationships, their relative lack of interpretability may limit adoption, particularly when performance improvements are minimal.

This study also highlights the importance of aligning model performance with clinical priorities. Adjusting the classification threshold from the default value of 0.5 to 0.3 demonstrated a clear trade-off between sensitivity and specificity. Lower thresholds improved the model's ability to identify high-risk patients but resulted in a higher number of false positive predictions. In many healthcare settings, this trade-off may be acceptable, as the consequences of failing to identify high-risk patients often outweigh the burden of additional precautionary interventions (Sharda et al., 2025). This flexibility underscores the value of predictive models as adaptable components within clinical decision support systems, where thresholds can be tuned based on operational capacity and risk tolerance.

Despite these strengths, several important considerations should be acknowledged when applying the results to real-world settings. The use of retrospective data from a single health system may limit the generalizability of the findings, and variables such as discharge disposition may not be fully reliable in real-time applications prior to patient discharge. Although structured EHR data are increasingly used for predictive analytics, they remain vulnerable to biases, inconsistencies, and missingness that may influence model performance.

Building on these findings, future research could expand this work in several important ways. Incorporating unstructured data sources, such as provider notes, may enhance predictive performance by capturing important clinical context that is often lacking in structured variables (Pandey et al., 2025). Additionally, integrating social determinants of health data could improve the identification of high-risk

patients and provide insight into disparities in readmission outcomes. Finally, the validation of these models in real-time clinical settings would further strengthen their practical applicability, while expanding the dataset across multiple health systems could improve generalizability.

Collectively, the results of this study suggest that while ML models offer modest improvements in predictive performance, LR remains a highly effective and practical approach in readmission prediction. The balance between interpretability, performance, and clinical usability is a critical consideration in healthcare analytics, and in this context, traditional statistical models may continue to provide the best value for real-world implementations.

RECOMMENDATIONS

Based on the findings of this study, several recommendations can be made for healthcare organizations seeking to leverage predictive analytics to identify patients at high risk of 30-day readmission. First, organizations should consider implementing an LR-based risk prediction model within their EHR systems. Given its comparable performance to more complex modeling approaches and its advantage in interpretability, LR offers a practical and efficient solution for real-time clinical decision support.

Second, predictive models should be integrated into clinical workflows as decision support tools that inform discharge planning and care coordination efforts. High risk scores generated during hospitalization – particularly as discharge approaches – should be reviewed for clinical appropriateness prior to initiating targeted interventions such as enhanced discharge education, medication reconciliation, or post-discharge follow-up. Embedding these tools within existing workflows ensures that model outputs are easily accessible, actionable, and effectively utilized by care teams.

Third, organization should consider the adoption of flexible classification thresholds based on clinical priorities and resource availability. In this study, lowering the prediction threshold from 0.5 to 0.3 substantially increased the identification of high-risk patients, though it also resulted in a higher number of false positives. Healthcare systems should tailor thresholds to align with operational capacity and tolerance for false positives, ensuring that interventions remain both effective and sustainable.

Finally, ongoing monitoring and evaluation of model performance are essential. As healthcare landscapes evolve – including changes in patient populations, clinical practices, and operational workflows – predictive models must be regularly updated and recalibrated to maintain accuracy and clinical relevance. The establishment of governance processes for model oversight are also recommended to ensure that predictive tools remain reliable, effective, and aligned with organizational goals.

CONCLUSION

This study examined the use of patient-level EHR data to predict 30-day hospital readmission risk and compared the performance of LR and ML models. The findings demonstrate that meaningful predictive capability can be achieved using data available during hospitalization, supporting the early identification of high-risk patients prior to discharge. Variables related to prior healthcare utilization, discharge disposition, and patient risk scores consistently emerged as the most influential predictors across all models, highlighting the importance of patient complexity and care transitions in readmission risk.

While ML models, particularly Gradient Boosting, achieved slightly higher predictive performance over LR, these gains were modest and not statistically significant. This suggests that increased model complexity does not necessarily result in meaningful improvements in predictive accuracy for this application. Instead, LR offers a practical and effective approach that balances performance with interpretability and ease of integration into clinical workflows.

Overall, the results of this study indicate that predictive models can serve as valuable tools for supporting discharge planning and care coordination efforts. By enabling the early identification of high-risk patients, healthcare organizations can implement targeted interventions aimed at reducing avoidable readmissions. The ability to adjust classification thresholds further enhances the adaptability of these models to align with organizational priorities and resource constraints. As the U.S. healthcare system continues to shift toward value-based care, predictive analytics will play an increasingly important role in improving patient outcomes, enhancing care delivery, and managing costs.

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